FEA Project 1

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Material Parameters:

Mooney-Rivlin material parameters:

To calculate material parameters A_{10}, A_{01} in a Mooney-Rivlin material from nominal stress data it is imperative to first understand the different conditions imposed by different tests on the deformation gradient, and how this relationship affects the Strain Energy Density Function and nominal stress function.

$$F = \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{bmatrix}, \qquad J = 1$$

$$J_1 = \lambda_1^2 + \lambda_2^2 + \lambda_3^2$$

$$J_2 = \lambda_1^{-2} + \lambda_2^{-2} + \lambda_3^{-2}$$

$$U = A_{10}(J_1 - 3) + A_{01}(J_2 - 3)$$

$$P = \frac{dU}{d\lambda}$$

In this study we examine nominal (First Piola-Kirchoff) uniaxial, biaxial and planar shear stress.

Uniaxial:

In uniaxial conditions, the first principle stretch $\lambda_1 = \lambda = e + 1$, where e is the measured strain. Because the main stretch in uniaxial is made in the first principal direction, and because of the incompressibility constrain ($J = 1 = \lambda_1\lambda_2\lambda_3$)

$$\lambda_1 = \lambda = e + 1, \qquad \lambda_2 = \lambda_3 = \lambda^{-1/2}$$

$$J_1 = \lambda^2 + 2\lambda^{-1}, \qquad J_2 = \lambda^{-2} + 2\lambda$$

$$U = A_{10}(J_1 - 3) + A_{01}(J_2 - 3)$$

$$= A_{10}(\lambda^2 + 2\lambda^{-1} - 3) + A_{01}(\lambda^{-2} + 2\lambda - 3)$$

$$P = \frac{dU}{d\lambda} = A_{10}(2\lambda - 2\lambda^{-2}) + A_{01}(-2\lambda^{-3} + 2)$$

Since we are given the value of P and the value of e, we can easily run a least squares equation solver to solve for A_{10} and A_{01}. Similar steps are followed to find Biaxial and Planar Shear Stress:

Biaxial:

In equal biaxial conditions, the first principle and the second principle stretch equal each other $\lambda_1 = \lambda_2 = \lambda = e + 1$, where e is the measured strain. Because of the incompressibility constrain ($J = 1 = \lambda_1\lambda_2\lambda_3$)

$$\lambda_1 = \lambda_2 = \lambda = e + 1, \qquad \lambda_3 = \lambda^{-2}$$

$$J_1 = 2\lambda^2 + \lambda^{-4}, \qquad J_2 = 2\lambda^{-2} + \lambda^4$$

$$U = A_{10}(J_1 - 3) + A_{01}(J_2 - 3)$$

$$= A_{10}(2\lambda^2 + \lambda^{-4} - 3) + A_{01}(2\lambda^{-2} + \lambda^4 - 3)$$

$$P = \frac{1}{2}\frac{dU}{d\lambda} = \frac{1}{2}A_{10}(4\lambda - 4\lambda^{-5}) + A_{01}(-4\lambda^{-3} + 4\lambda^3) =$$

$$2\big(A_{10}(\lambda - \lambda^{-5}) + A_{01}(\lambda^3 - \lambda^{-3})\big)$$

Planar Shear:

In the case of pure shear, the principle stretches can be written in terms of the measured strain as:

$$\lambda_1 = \lambda = e + 1, \qquad \lambda_2 = 1, \qquad \lambda_3 = \frac{1}{\lambda}$$

Due to the incompressibility constraint. Which sets the deviatoric invariants equal to:

$$J_1 = \lambda^2 + 1 + \lambda^{-2}, \qquad J_2 = \lambda^2 + 1 + \lambda^{-2}$$

$$U = A_{10}(\lambda^2 + \lambda^{-2} - 2) + A_{01}(\lambda^2 + \lambda^{-2} - 2)$$

$$P = \frac{dU}{d\lambda} = A_{10}(2\lambda - 2\lambda^{-3}) + A_{01}(2\lambda - 2\lambda^{-3}) =$$

$$2\big(A_{10}(\lambda - \lambda^{-3}) + A_{01}(\lambda - \lambda^{-3})\big)$$

System of Equations:

Since we are given the nominal stress results from all three tests, we can build a system of equations and use a least squares solver to find A_{01}& A_{10}

(I used scipy.optimize.least_squares from the scipy library in python)

$$Eq1 = A_{10}(2\lambda - 2\lambda^{-2}) + A_{01}(-2\lambda^{-3} + 2) - P_{axial}$$

$$Eq2 = A_{10}(2\lambda - 2\lambda^{-5}) + A_{01}(2\lambda^{3} - 2\lambda^{-3}) - P_{biaxial}$$

$$Eq3 = A_{10}(2\lambda - 2\lambda^{-3}) + A_{01}(2\lambda - 2\lambda^{-3}) - P_{shear}$$

Plugging these equations into the aforementioned least squares solver returned:

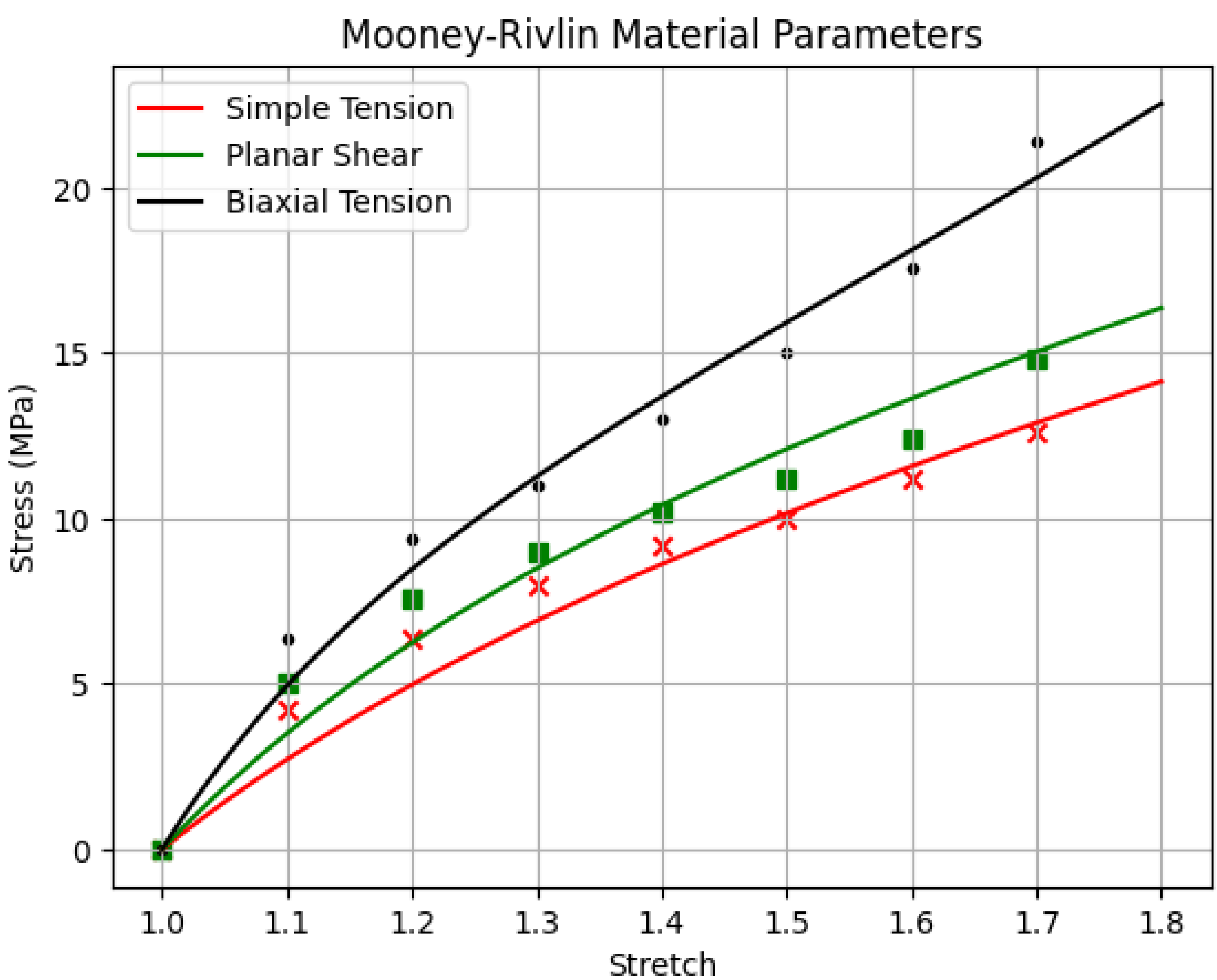
$$A_{10} = 4.38649768, \qquad A_{01} = 0.63733341$$

While the material evaluator in Abaqus returned:

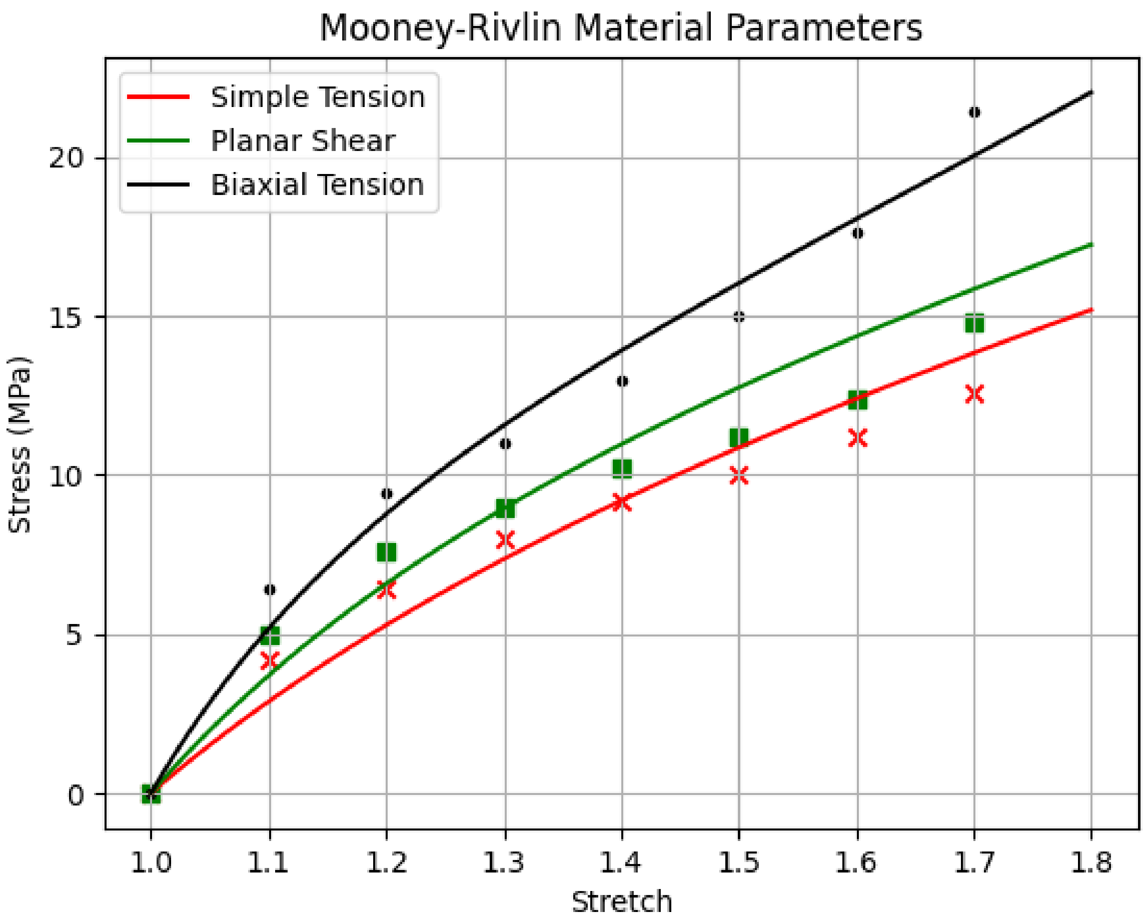
$$A_{10} = 4.8397, \qquad A_{01} = 0.45122$$

They are similar albeit not exact. I trust this difference to be in how the least squares function operates in python versus in abaqus.

Plotting the python fitted curves against the data returns this plot:



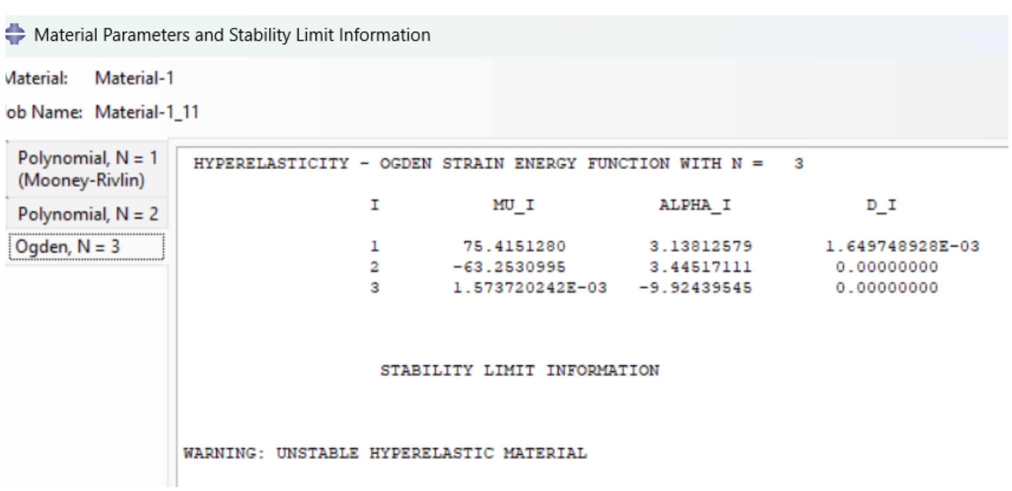
Plotting the abaqus coefficients returns this plot:



Both plots fit the data reasonably enough.

Ogden Material Parameters:

From the Abaqus material evaluation


Material Parameters and Stability Limit Information

Material: Material-1

Job Name: Material-1_11

Polynomial, N = 1 (Mooney-Rivlin)

Polynomial, N = 2

Ogden, N = 3

HYPERELASTICITY - OGDEN STRAIN ENERGY FUNCTION WITH N = 3

I	MU_I	ALPHA_I	D_I
1	75.4151280	3.13812579	1.649748928E-03
2	-63.2530995	3.44517111	0.00000000
3	1.573720242E-03	-9.92439545	0.00000000

STABILITY LIMIT INFORMATION

WARNING: UNSTABLE HYPERELASTIC MATERIAL

$$\alpha_1 = 3.13813, \qquad \alpha_2 = 3.44517, \qquad \alpha_3 = -9.92439$$

$$\mu_1 = 75.4151280, \qquad \mu_2 = -63.2530995, \qquad \mu_3 = 1.573720242 * 10^{-3}$$

$$W(\lambda_1, \lambda_2, \lambda_3) = \sum_{i=1}^{N} \frac{\mu_i}{\alpha_i} \left(\lambda_1^{\alpha_i} + \lambda_2^{\alpha_i} + \lambda_3^{\alpha_i} - 3\right)$$

$$P = \frac{dW}{d\lambda}$$

From here a similar process to the Mooney-Rivlin material is taken, but this time the parameters are known and the invariants are not needed.

Uniaxial:

$$\lambda_1 = \lambda = e + 1, \qquad \lambda_2 = \lambda_3 = \lambda^{-1/2}$$

$$W(\lambda_1, \lambda_2, \lambda_3) = \sum_{i=1}^{N} \frac{\mu_i}{\alpha_i} \left(\lambda_1^{\alpha_i} + \lambda_2^{\alpha_i} + \lambda_3^{\alpha_i} - 3\right)$$

$$= \sum_{i=1}^{3} \frac{\mu_i}{\alpha_i} \left(\lambda^{\alpha_i} + 2\lambda^{-\alpha_i/2} - 3\right)$$

$$P = \frac{dW}{d\lambda} = \sum_{i=1}^{3} \frac{\mu_i}{\alpha_i} \left(\alpha_i \lambda^{\alpha_i - 1} - \alpha_i \lambda^{-\frac{\alpha_i - 2}{2}}\right)$$

$$P = \sum_{i=1}^{3} \mu_i \left(\lambda^{\alpha_i - 1} - \lambda^{-\frac{\alpha_i - 2}{2}}\right)$$

Biaxial:

$$\lambda_1 = \lambda_2 = \lambda = e + 1, \qquad \lambda_3 = \lambda^{-2}$$

$$W(\lambda_1, \lambda_2, \lambda_3) = \sum_{i=1}^{N} \frac{\mu_i}{\alpha_i} \left(\lambda_1^{\alpha_i} + \lambda_2^{\alpha_i} + \lambda_3^{\alpha_i} - 3\right)$$

$$= \sum_{i=1}^{3} \frac{\mu_i}{\alpha_i} (2\lambda^{\alpha_i} + \lambda^{-2\alpha_i} - 3)$$

$$P = \frac{1}{2}\frac{dW}{d\lambda} = \sum_{i=1}^{3} \frac{\mu_i}{2\alpha_i}\left(2\alpha_i \lambda^{\alpha_i - 1} - 2\alpha_i \lambda^{-2\alpha_i - 1}\right)$$

$$P = \sum_{i=1}^{3} \mu_i\left(\lambda^{\alpha_i - 1} - \lambda^{-2\alpha_i - 1}\right)$$

Shear:

$$\lambda_1 = \lambda = e + 1, \qquad \lambda_2 = 1, \qquad \lambda_3 = \frac{1}{\lambda}$$

$$W(\lambda_1, \lambda_2, \lambda_3) = \sum_{i=1}^{N} \frac{\mu_i}{\alpha_i}\left(\lambda_1^{\alpha_i} + \lambda_2^{\alpha_i} + \lambda_3^{\alpha_i} - 3\right)$$

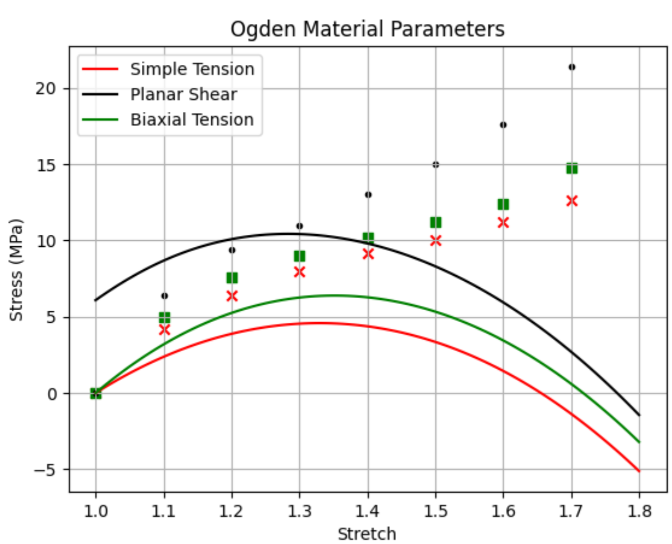
$$= \sum_{i=1}^{3} \frac{\mu_i}{\alpha_i}\left(\lambda^{\alpha_i} + \lambda^{-\alpha_i} - 2\right)$$

$$P = \frac{dW}{d\lambda} = \sum_{i=1}^{3} \frac{\mu_i}{\alpha_i}\left(\alpha_i \lambda^{\alpha_i - 1} - \alpha_i \lambda^{-\alpha_i - 1}\right)$$

$$P = \sum_{i=1}^{3} \mu_i\left(\lambda^{\alpha_i - 1} - \lambda^{-\alpha_i - 1}\right)$$

Plotting:

Plotting the equations for the first Piola-Kirchoff stress in Python with the provided alpha and mu values results in this graph:

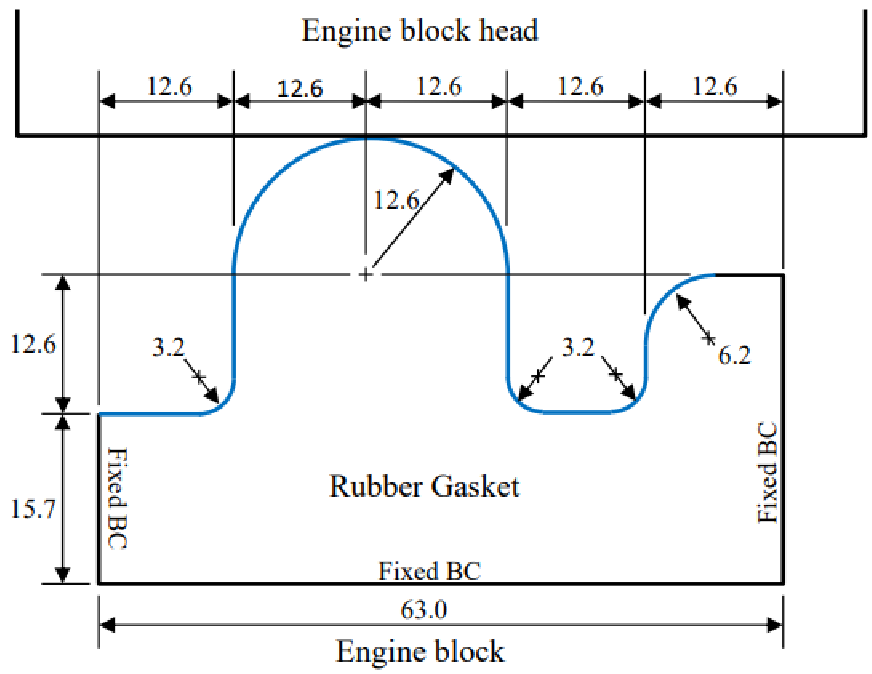


As seen in the graph, Ogden materials do a very poor job of fitting our test results.

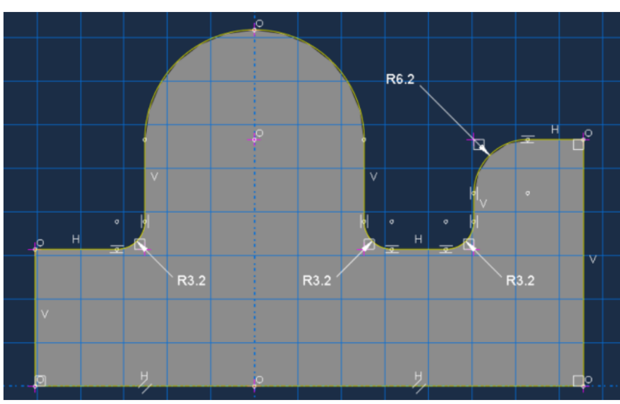
Because of this we will progress with the Mooney-Rivlin Material for our Abaqus analysis.

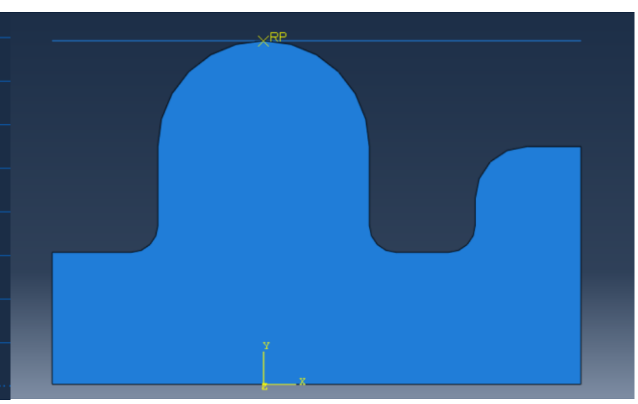
Abaqus Analysis:

Basic Model:



There are two parts in this model: The gasket, and the Engine Block head. The Rubber Gasket is made of the Mooney-Rivlin material we evaluated and the Engine Block Head, since it is much stiffer than the gasket can be modelled as an analytical rigid body.



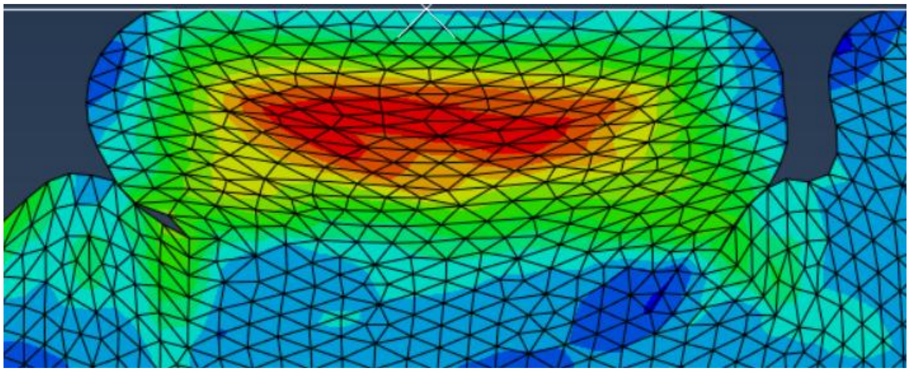


A self-contact interaction is set between the gasket and its surface, and another interaction couples the engine head to the gasket. When a boundary condition is tied to the time step that moves the engine head down vertically, the gasket compresses with it.

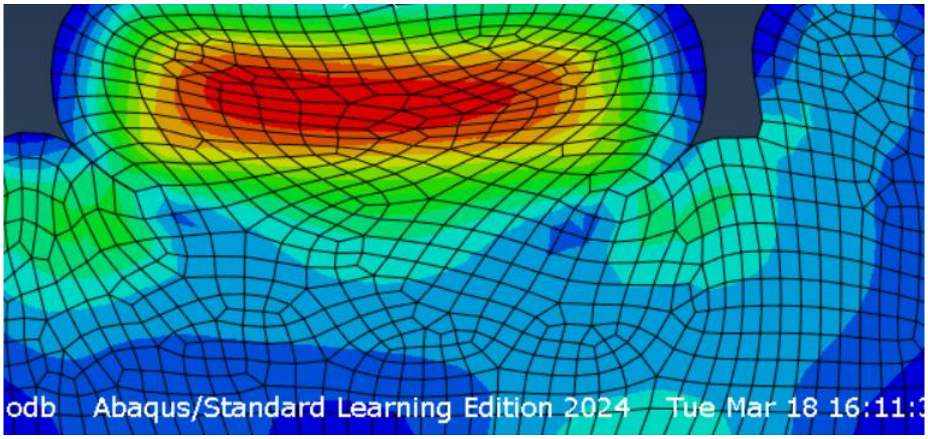
Mesh Type:

I decided to go with a quadrilateral mesh because I thought it would make the surface smoother and less jagged, since surface contact is very important in this model.

Triangular



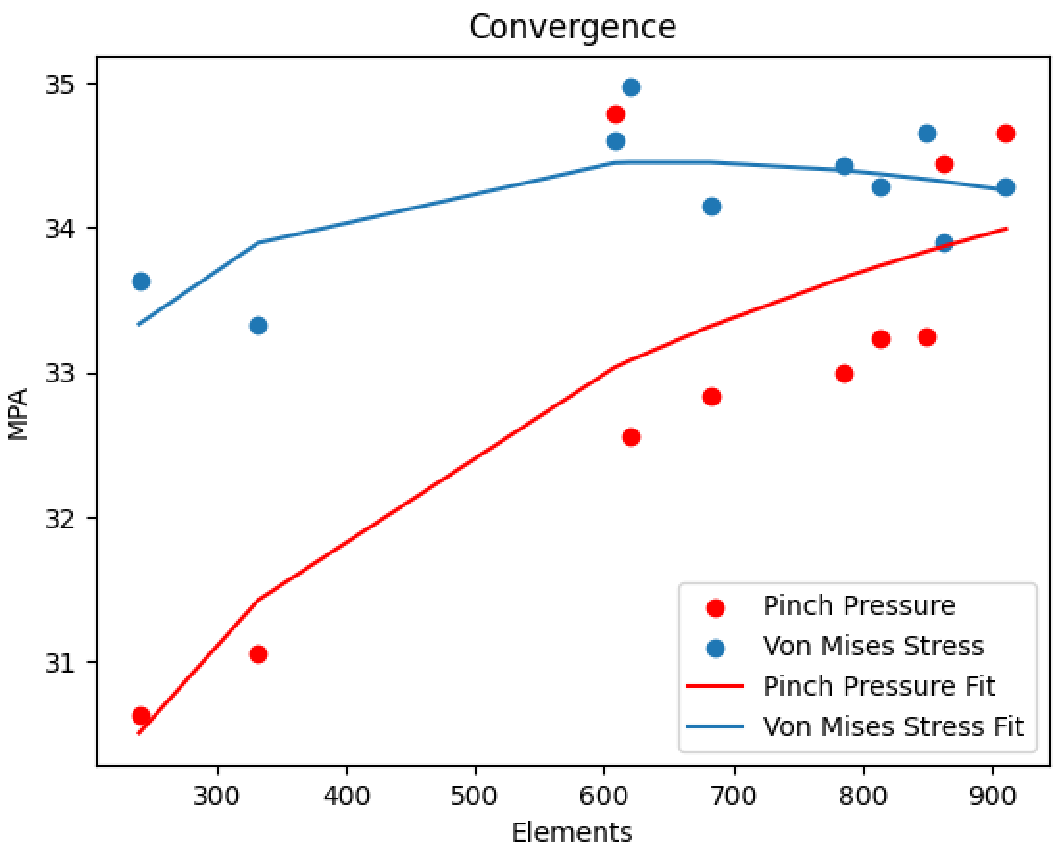
Quadrilateral



Observe the gap in surface to surface contact in the on the left side of the model, which is not how the surface would react. The quadrilateral deformation in general has a smoother surface.

Mesh Convergence:

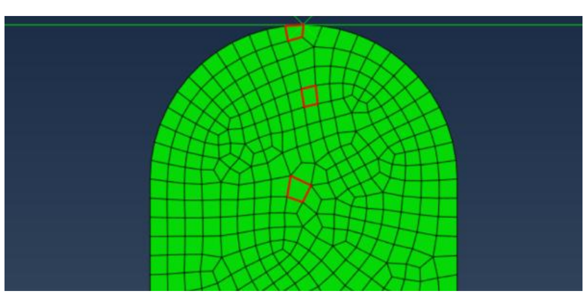
In order to test the best mesh size to use for quadrilateral elements in this model a convergence analysis was performed. Different amounts of elements were generated in the model and the contact pressure between the surfaces on the right side and the maximum Von Mises stress were recorded for each element size.

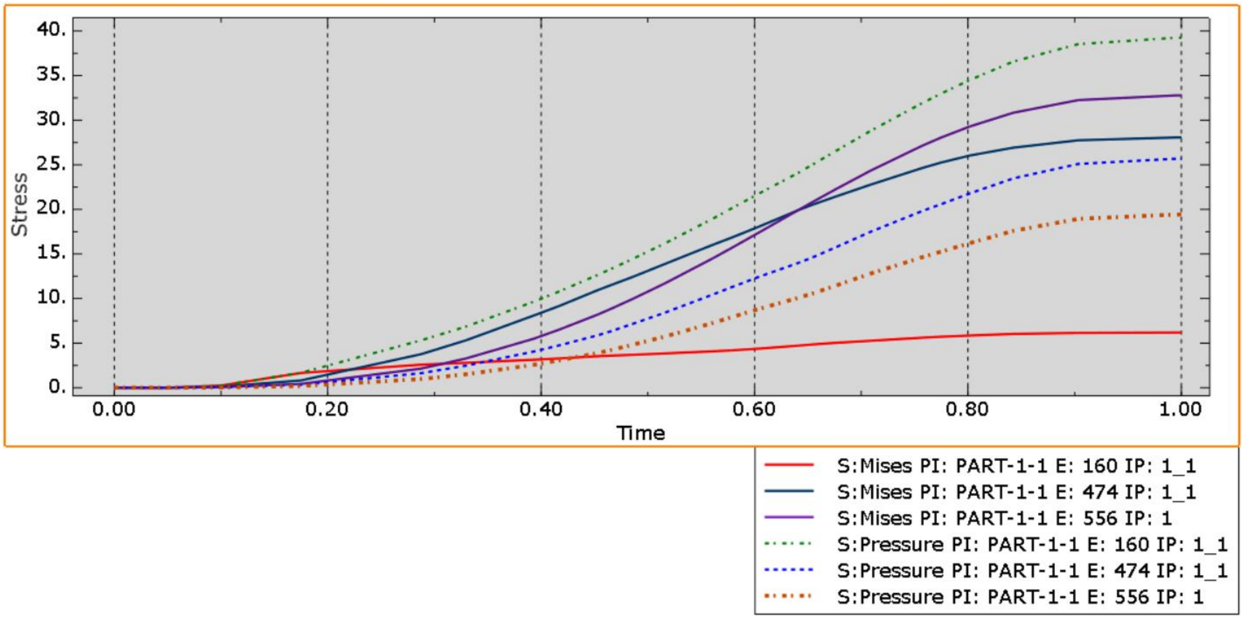


It seems that the model begins to converge at around 600 elements for the von mises stress and a bit higher than that for the contact pressure. (Unfortunately, the learning edition of Abaqus restricts me to 1000 nodes so I couldn't model anything higher). While the model does seem to converge at 600 elements (element size of 2), I will be using a slightly smaller element size for the rest of the analysis (element size of 1.5).

Pressure and Von-Mises Stress Analysis:

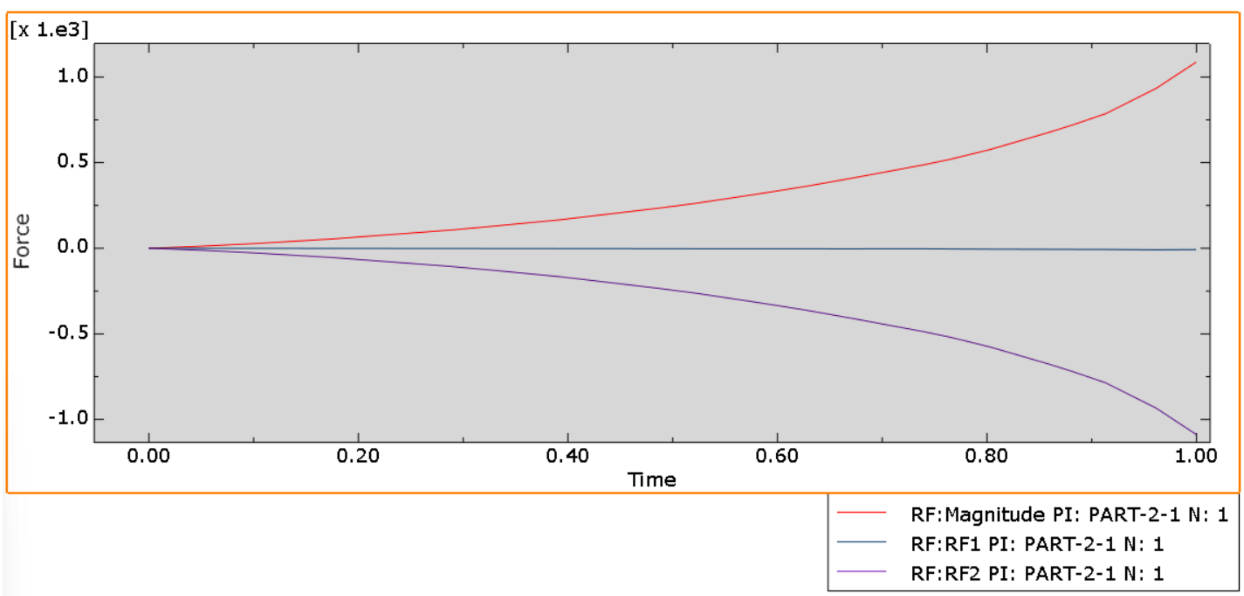
Stress and Von-Mises graphs were generated for the below nodes:





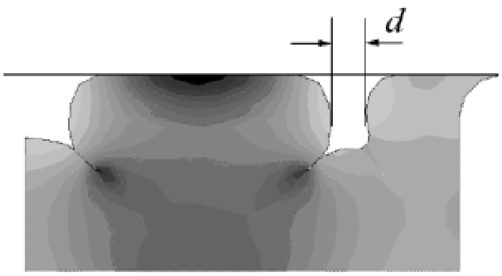
Initially the elements closer to the contact between the gasket and the engine block have a higher Von Mises stress, but as time progresses and the engine block head approaches a peak compression of 12.6 mm, the elements further away from the contact point have the higher Von Mises stress. Conversely the elements closest to the contact point always have the higher pressure.

Load vs. Time of Engine Block Head



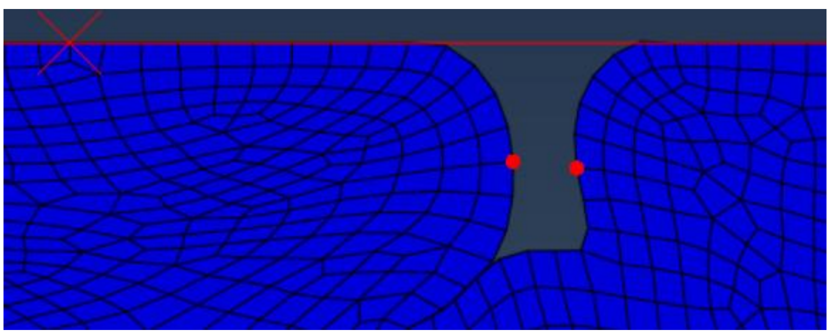
The reaction force in the engine block head is equal to reaction force experienced by the reference point. The reaction force is only in the y direction and its maximum value is 1.088*10^3 N.

Reduction of the Gap Size:



Original Gap Size:

When measured with the Abaqus the original deformed gap size d is equal to 2.86 mm, with the undeformed gap size being 12.6 mm.



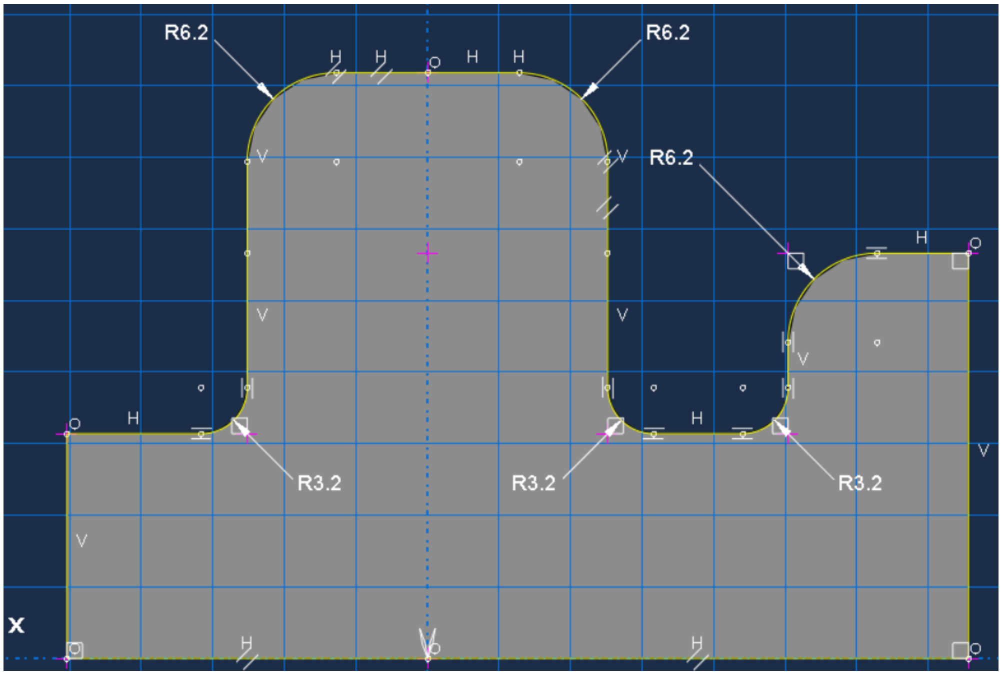
The goal is to minimize this gap while keeping the contact pressure as uniform as possible.

New Proposed Design:

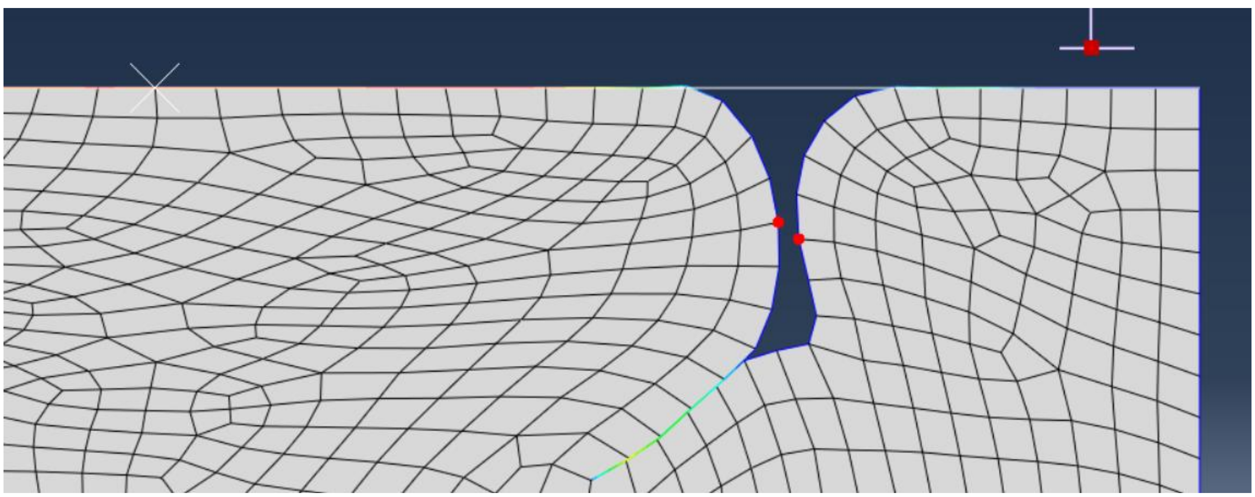
Key Design Constraints:

A trivial way to reduce the deformed gap size would be to reduce undeformed gap size. This feels cheap and insincere. Lets assume that the undeformed gap size is a critical component of the original design and its reduction would be catastrophic to the performance of the part.

A better way to approach this problem would be to increase the material volume undergoing compression by the engine block head. This can be done by increasing the height of the gasket or widening the gasket head. Increasing the height of the gasket would mean increasing compression distance, so the design I went with increased material volume of the gasket head by putting a filleted block as the head as opposed to a semi-circle.



In the analysis I use the same mesh type and mesh size (quadrilateral 1.5mm)



D was reduced to 0.74035 mm with this design compared to the original designs 2.86 mm. As mentioned previously the same undeformed distance of 12.6 mm was kept in the design.

The original contact pressure on the right surface contact interface was

```
 Part Instance      Node ID  Attached elements          CPRESS
-------------------------------------------------------------------
    PART-1-1           46       Elem59:f1:          6.91957
    PART-1-1           46      Elem454:f4:          6.91957
    PART-1-1           47       Elem59:f1:          22.2444
    PART-1-1           47      Elem118:f4:          22.2444
    PART-1-1           48      Elem121:f3:          16.2938
    PART-1-1           48      Elem454:f4:          16.2938
    PART-1-1           45      Elem118:f4:          12.9023
    PART-1-1           45      Elem352:f2:          12.9023
    PART-1-1           49      Elem121:f3:          9.95986
    PART-1-1           49      Elem122:f3:          9.95986
    PART-1-1          265      Elem117:f4:          7.93286
    PART-1-1          265      Elem352:f2:          7.93286
```

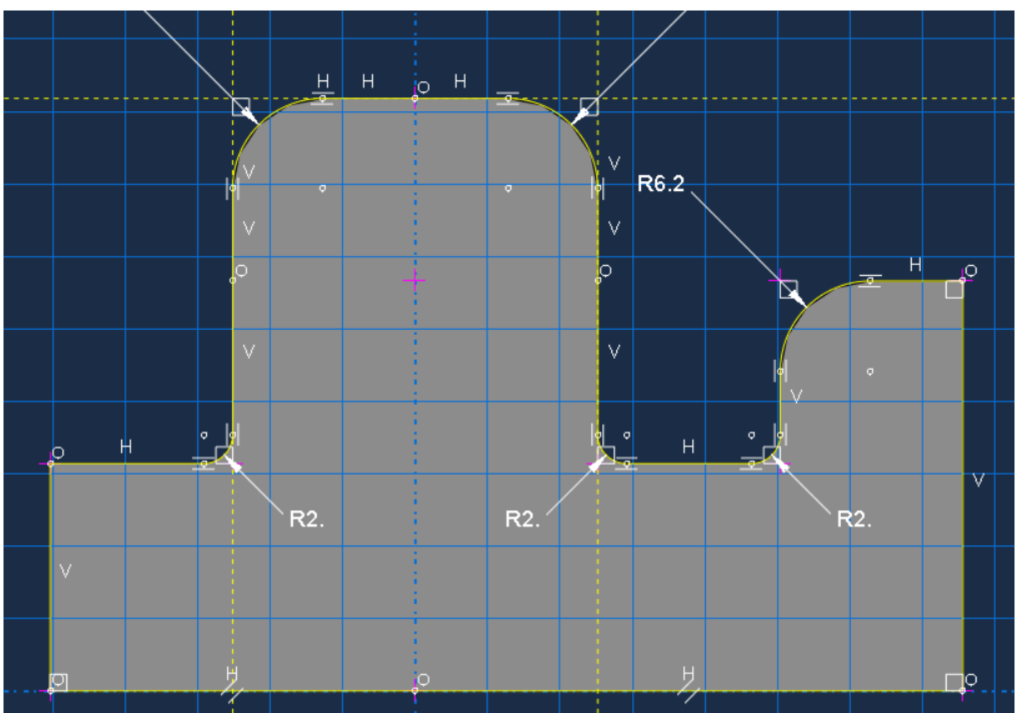
And the new contact surface pressure is:

```
 Part Instance      Node ID  Attached elements          CPRESS
-------------------------------------------------------------------
    PART-1-1           46       Elem58:f1:          6.25921
    PART-1-1           46      Elem450:f4:          6.25921
    PART-1-1           47       Elem58:f1:          30.6544
    PART-1-1           47      Elem113:f4:          30.6544
    PART-1-1           48      Elem116:f3:          27.4411
    PART-1-1           48      Elem450:f4:          27.4411
    PART-1-1           49      Elem116:f3:          19.7806
    PART-1-1           49      Elem117:f3:          19.7806
    PART-1-1          272      Elem117:f3:          17.0398
    PART-1-1          272      Elem445:f1:          17.0398
    PART-1-1          271      Elem112:f4:          17.5521
    PART-1-1          271      Elem354:f2:          17.5521
    PART-1-1           44      Elem112:f4:          13.0532
    PART-1-1           44      Elem353:f4:          13.0532
    PART-1-1           50      Elem355:f3:          10.4141
    PART-1-1           50      Elem445:f1:          10.4141
    PART-1-1          270       Elem33:f4:          3.34875
    PART-1-1          270      Elem353:f4:          3.34875
```

The new contact pressure has a range of $[6.2591 \quad 30.6544]$ while the old has a range of $[6.91957 \quad 22.2444]$. The range increased by almost 9 MPa, but is to be expected. Further adjustments must be made.

Reducing the fillet size on the modified part from 3.2 to 2mm gives a smaller range of contact pressure values $[6.30664 \quad 24.0779]$ while preserving a distance of 0.74 mm.

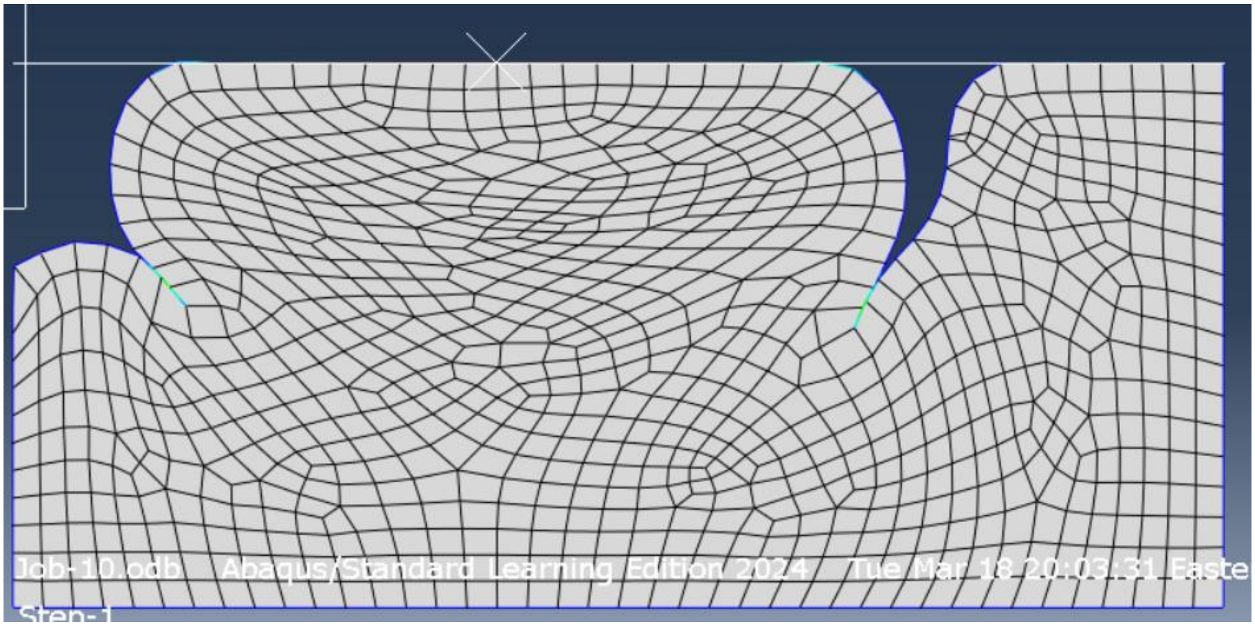
Part Instance	Node ID	Attached elements	CPRESS
PART-1-1	16	Elem15:f3:	10.9384
PART-1-1	16	Elem135:f2:	10.9384
PART-1-1	109	Elem20:f2:	24.0779
PART-1-1	109	Elem135:f2:	24.0779
PART-1-1	110	Elem20:f2:	20.1756
PART-1-1	110	Elem21:f2:	20.1756
PART-1-1	111	Elem21:f2:	18.3909
PART-1-1	111	Elem99:f2:	18.3909
PART-1-1	112	Elem99:f2:	12.0235
PART-1-1	112	Elem144:f2:	12.0235
PART-1-1	106	Elem63:f4:	6.30664
PART-1-1	106	Elem64:f2:	6.30664



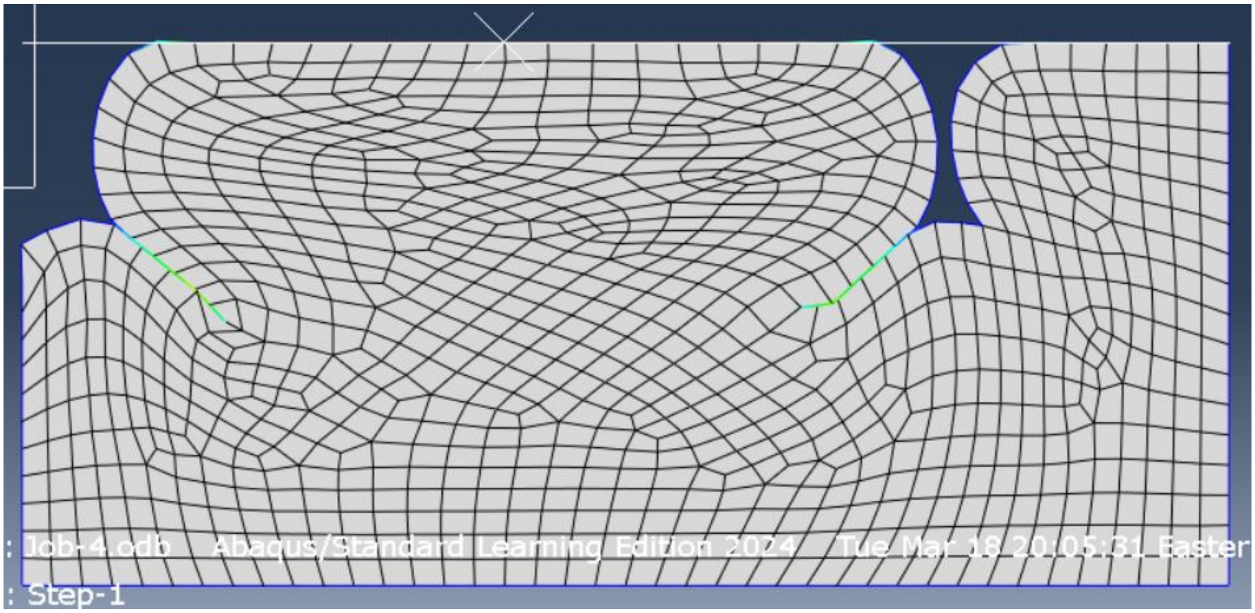
This is the final design I will go with. Further reductions of radius break the model and increasing the radius increases total contact area and pressure for this geometry, while decreasing distance.

When increasing the fillet on the original design, contact length decreases as well as pressure comparatively to the new design.

Old with fillets of 6.3mm



New with fillets of 2 mm



Code:

```
# Mooney-Rivlin material parameter calculations

# TEST DATA
from scipy.optimize import least_squares
import numpy as np
import matplotlib.pyplot as plt
plt.close('all')
E = [0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7]  # Strain
s_xx = np.array([0, 4.2, 6.4, 8, 9.2, 10, 11.2, 12.6])  # Simple tension (MPa)
tau = np.array([0, 5, 7.6, 9, 10.2, 11.2, 12.4, 14.8])  # Planar shear (MPa)
s_xx_yy = np.array([0, 6.4, 9.4, 11, 13, 15, 17.6, 21.4])  # Biaxial tension
(MPa)

L = [e + 1 for e in E]
stretch = np.array(L)

# Mooney-Rivlin Material Parameters Equations
def EQ(A):
    term1 = (A[0] * 2 * (stretch - stretch**-2) + A[1] * 2 * (1 - stretch**-3)) -
s_xx
    term2 = (A[0] * 2 * (stretch - stretch**-3) + A[1] * 2 * (stretch -
stretch**-3)) - tau
    term3 = (A[0] * 2 * (stretch - stretch**-5) + A[1] * 2 * (stretch**3 -
stretch**-3)) - s_xx_yy

    return np.concatenate((term1, term2, term3))

#from the textbook page 225 - 227



initial_guess =np.random.rand(3)
results = least_squares(EQ, initial_guess, method= 'trf',bounds=(0, np.inf),
verbose=1)
print(results.x)
A10 = results.x[0]
A01 = results.x[1]

#plotting the results
l = np.linspace(1,1.8,100)

polynomial = A10*2*(l - l**-2) + A01*2*(1 - l**-3)
phantom = A10 * 2 * (l - l**-3) + A01 * 2 * (l - l**-3)
```

```
bloodmoon = A10 * 2 * (l - l**-5) + A01 * 2 * (l**3 - l**-3)
plt.plot(l, polynomial, color ='r')
plt.plot(l, phantom, color='g')
plt.plot(l, bloodmoon, color='black')
plt.scatter(L, s_xx, color='r', marker='x')
plt.scatter(L, tau, color='g', marker='s')
plt.scatter(L, s_xx_yy, color='black', marker='.')
plt.xlabel('Stretch')
plt.ylabel('Stress (MPa)')
plt.legend([ 'Simple Tension', 'Planar Shear', 'Biaxial Tension'])
plt.title('Mooney-Rivlin Material Parameters')
plt.grid()
plt.show()

A10 = 4.8397
A01 = 0.45122
polynomial = A10*2*(l - l**-2) + A01*2*(1 - l**-3)
phantom = A10 * 2 * (l - l**-3) + A01 * 2 * (l - l**-3)
bloodmoon = A10 * 2 * (l - l**-5) + A01 * 2 * (l**3 - l**-3)
plt.plot(l, polynomial, color ='r')
plt.plot(l, phantom, color='g')
plt.plot(l, bloodmoon, color='black')
plt.scatter(L, s_xx, color='r', marker='x')
plt.scatter(L, tau, color='g', marker='s')
plt.scatter(L, s_xx_yy, color='black', marker='.')
plt.xlabel('Stretch')
plt.ylabel('Stress (MPa)')
plt.legend([ 'Simple Tension', 'Planar Shear', 'Biaxial Tension'])
plt.title('Mooney-Rivlin Material Parameters')
plt.grid()
plt.show()

#Ogden Material Parameters Equations and Plotting; From textbook page 188-189

mu = np.array([75.4151280, -63.2530995 , 1.573720242E-03])
a = np.array([3.13812579, 3.44517111, -9.92439545])
P_axial = []
P_biaxial = []
P_shear = []
for i in range(2):
    #axial tension
    P= mu[i]* (l**(a[i]-1) -l**(-a[i]/2-1))
    P_axial.append(P)
    P = 1/2*mu[i]*(2*l**(a[i]-1) - l**(-2*a[i]-1))
    P_biaxial.append(P)
```

```
    P= mu[i]*(l**(a[i]-1) - l**(-a[i]-1))
    P_shear.append(P)
P_axial = np.sum(P_axial, axis=0)
P_biaxial = np.sum(P_biaxial, axis=0)
P_shear = np.sum(P_shear, axis=0)


plt.plot(l, P_axial, color='r')
plt.plot(l, P_biaxial, color='black')
plt.plot(l, P_shear, color='g')
plt.scatter(L, s_xx, color='r', marker='x')
plt.scatter(L, tau, color='g', marker='s')
plt.scatter(L, s_xx_yy, color='black', marker='.')
plt.xlabel('Stretch')
plt.ylabel('Stress (MPa)')
plt.legend([ 'Simple Tension', 'Planar Shear', 'Biaxial Tension'])
plt.title('Ogden Material Parameters')
plt.grid()
plt.show()
```

References:

[1] N.-H. Kim, Introduction to Nonlinear Finite Element Analysis. Springer, 2015.